6.4 Deployment view

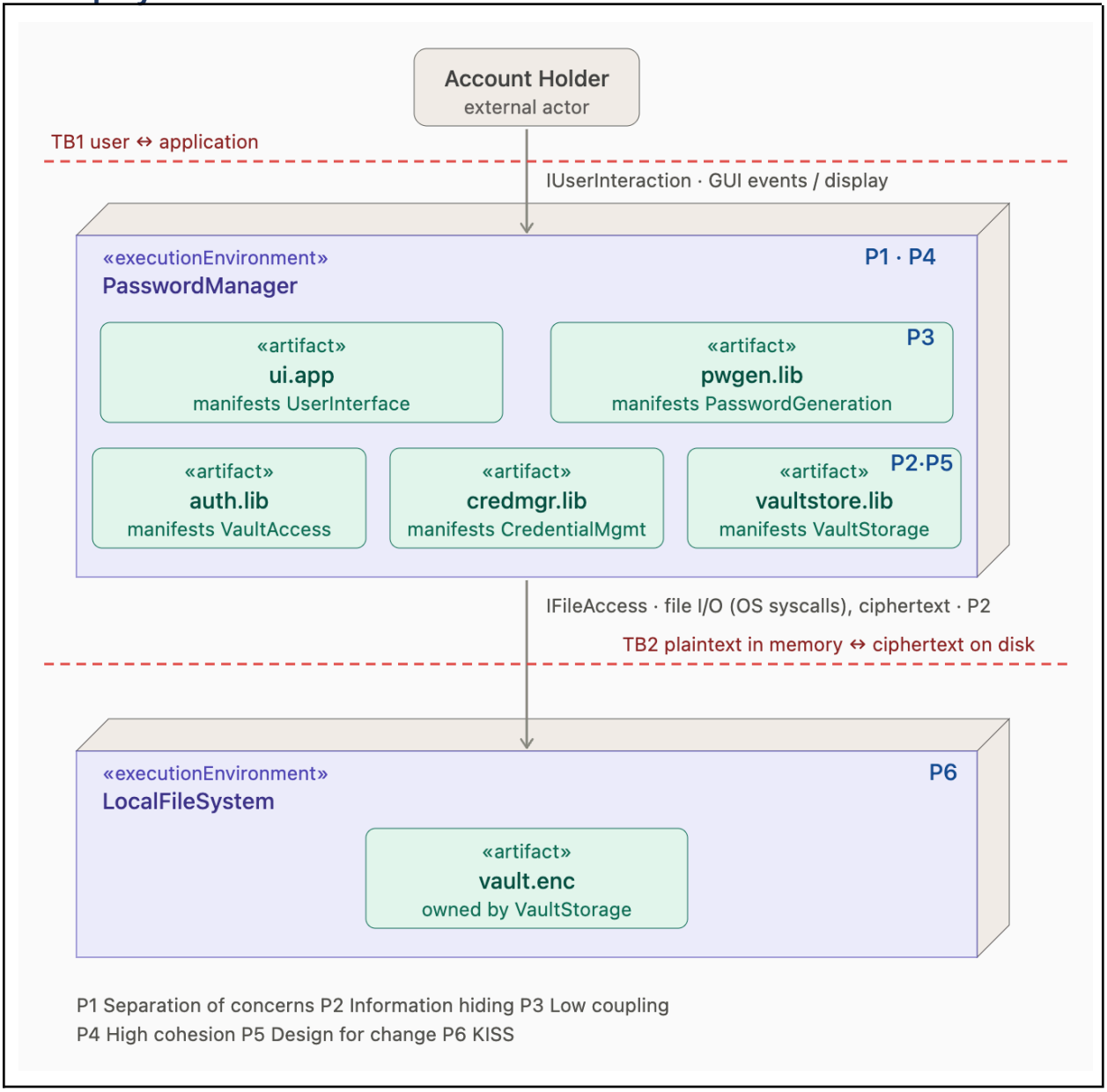


Figure 6.4 UML deployment diagram: nodes, execution environments, deployed artifacts, communication paths with protocols, trust boundaries

Node	What runs on it (artifacts, components)	Communicates with (protocol)	Trust zone
PasswordManager	**Execution Environment with UI.app and Pwgen.lib, auth.lib, credmgr.lib and vaultsore.lib**	**LocalFileSystem**	**TB1**

LocalStorage	Device - with vault.enc environment	N/A	TB2

6.5 From analysis to architecture

Analysis object (from 5.1 and 5.2)	Component that owns it (6.3)	Note (split, merged, external)
AccountHolder	User Interface	External - not part of the system
Vault	Vault Storage/Encryption	Split - Credentials in VaultStorage/Encryption. Lock status is in authentication/vault Access
Credential	Credential Management	Not split - in it's own component
PasswordCriteria	Password Generation	Merged with GeneratedPassword
GeneratedPassword	Password Generation	Merged with PasswordCriteria
GeneratorController	User Interface	Split: generation, conflict check go to Password Generation

6.6 Design principles applied

Principle	Where it shows in your design	Evidence (component, interface, ADR)
Separation of concerns	P1 - Only one artifact per responsibility in Password Manager	All components of 6.3 are represented via one artifact
Information hiding	P2 - Only vaultstore.lib has vaul.enc.	IStorge is the only way to reach it. IFileAccess is the only way into the file system.
Low coupling	P3 - wgen.lib has no dependency on the vault artifacts	PasswordGeneration only needs UserInteraction
High cohesion	P4 - Each artifact only has one component with a single responsibility	Each component handles something independant
Design for change	P5 - Encryption format is from	Easy to change because is

	vaultstore.lib	**modular so change mainly affects vaultstore.lib**
Keep it simple (KISS)	**Just two nodes**	**Deployment view**